TransKAN: Transfer Learning and Kolmogorov–Arnold Networks for Gunshot Wound Classification

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Telling an **entry** from an **exit** gunshot wound, and estimating the **shooting distance**, are core forensic tasks. They still rely on expert judgement and depend on lighting and imaging conditions at the scene. Deep CNNs have been tried, but conventional fully connected (MLP) heads struggle with the *irregular, noisy and imbalanced* decision boundaries of real forensic photographs. **TransKAN** keeps a pre-trained **ResNet-152** backbone and replaces its dense classifier with a **Kolmogorov–Arnold Network (KAN)** head, which has *learnable spline activations on every edge*. It is validated on **GuWID**, a new database of **2,551 real wound images** from Federal District Civil Police (PCDF) casework.

Real Forensic Evidence

⚠ *Contains real forensic imagery.*

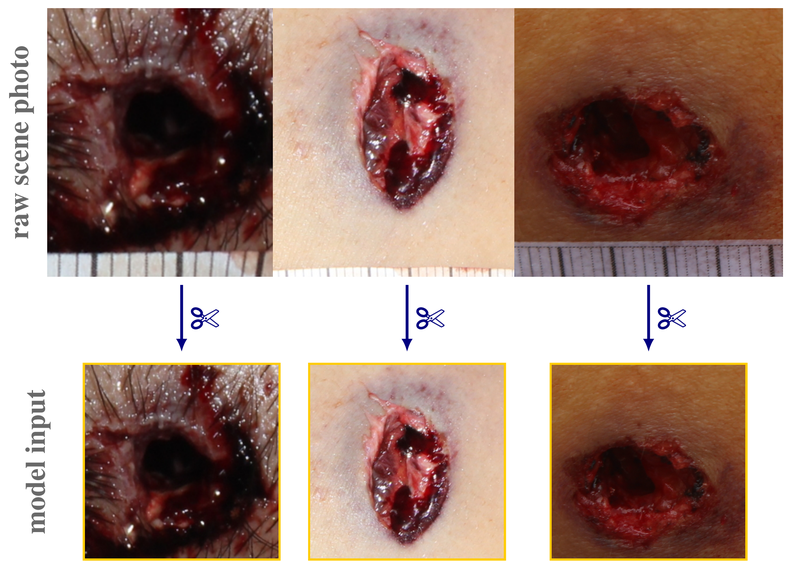


Figure 1: From crime-scene photograph to model input. Raw PCDF photographs (top; note the forensic ruler) are cropped in GIMP to a square region centred on the wound (bottom), then resized to 224×224. All three samples are entry wounds.

GuWID: Gunshot Wound Image Database

Built from records of **real homicide and suicide cases** examined by PCDF forensic experts (UnB Research Ethics Committee, protocol 4418221.9.0000.0030). Labels (wound type, anatomical location, shooting distance) were transcribed *verbatim* from the forensic pathologist's autopsy reports.

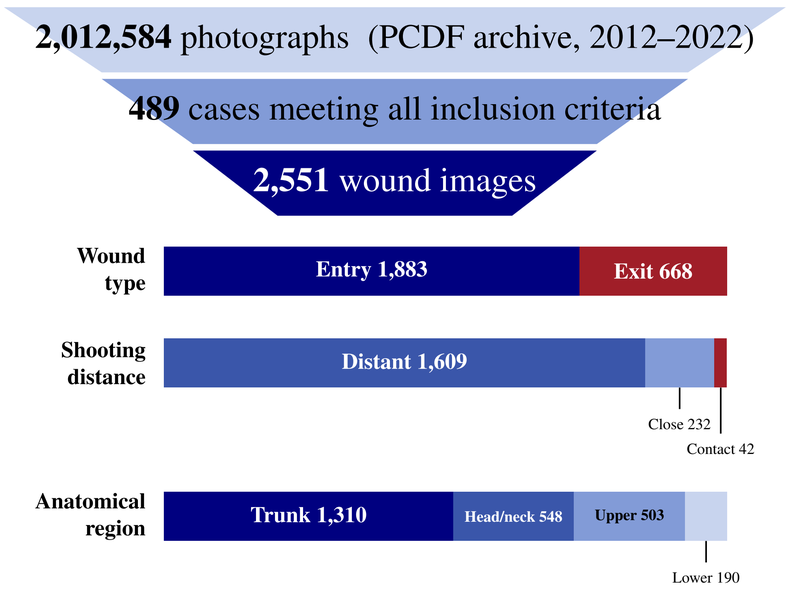


Figure 2: GuWID is heavily imbalanced, reflecting the real rarity of some injuries: exit wounds are 26% of the data and contact shots only 2.2% of entry wounds. 95.5% of images come from male victims, which matches the regional profile of firearm deaths.

TransKAN Architecture

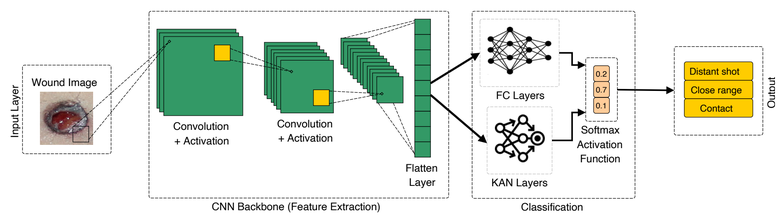
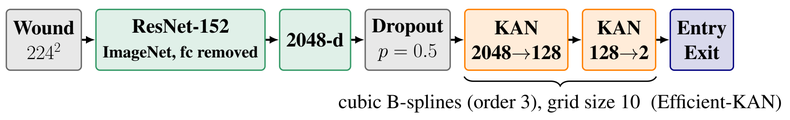


Figure 3: Pipeline. A CNN backbone extracts features that are flattened and passed to *either* a conventional FC head (baseline) *or* our KAN head. The two heads are separate configurations, trained and evaluated independently. They are drawn side by side only for comparison.



Why a KAN head? MLP vs. KAN

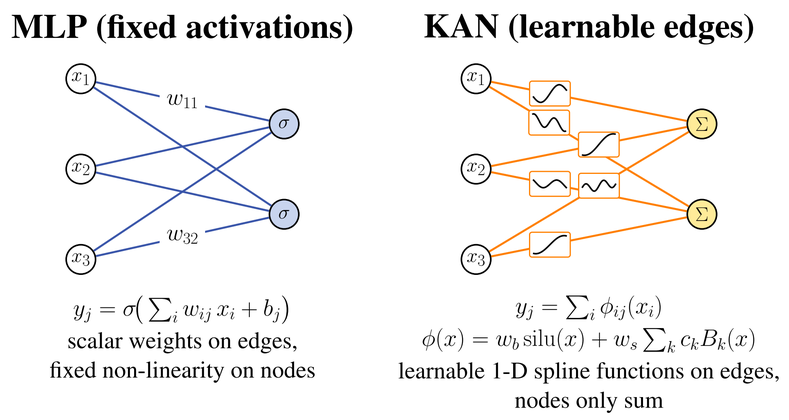
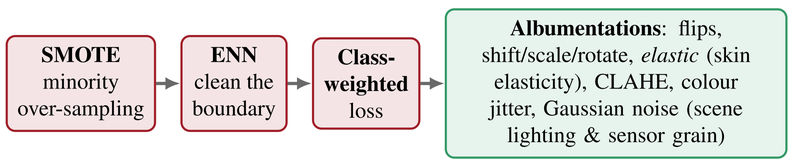


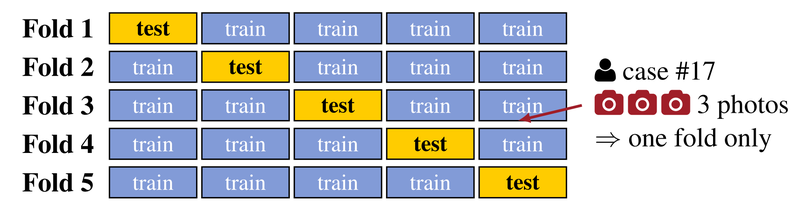
Figure 4: Edges learn their own activation. Each KAN edge is a B-spline ϕ_{ij} whose shape is learned. This gives the classifier the flexibility to fit irregular boundaries between morphologically similar wounds *without* extra depth, and only in the head, so the trainable parameter count stays modest on a few thousand images. The design is motivated by the Kolmogorov–Arnold representation theorem.

Imbalance Handling & Augmentation



All steps are **re-fit inside each training fold only**, so validation and test data are never touched.

Leakage-Free Evaluation

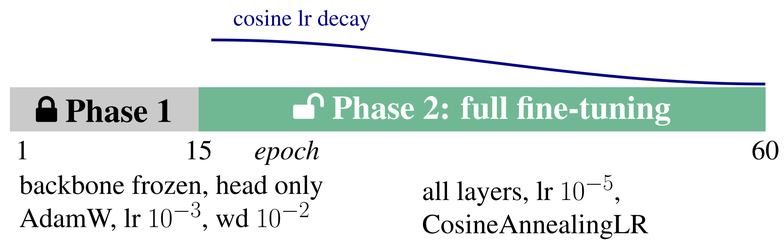


One victim may have several photographs of the *same* wound. We use **grouped, stratified 5-fold cross-validation by case ID**, so no wound appears in both training and test data. **All 12 baseline families were re-trained under exactly the same protocol**, and out-of-fold predictions are aggregated into the final metrics.

Take-home Messages

- Replacing only the **classifier head** (MLP→KAN) on a ResNet-152 improves precision, F1 and class separation (AUC) on real forensic images.
- **GuWID**: 2,551 authentic wound images, labelled from autopsy reports.
- **Case-grouped** cross-validation removes the photo-level leakage that inflates results.

Two-Phase Training



Results: Entry vs. Exit

macro F1	macro Precision	Accuracy	macro AUC
0.832	**0.847**	**0.871**	**0.893**
ResNet-152: 0.824	ResNet-152: 0.827	ResNet-152: 0.869	ResNet-152: 0.821

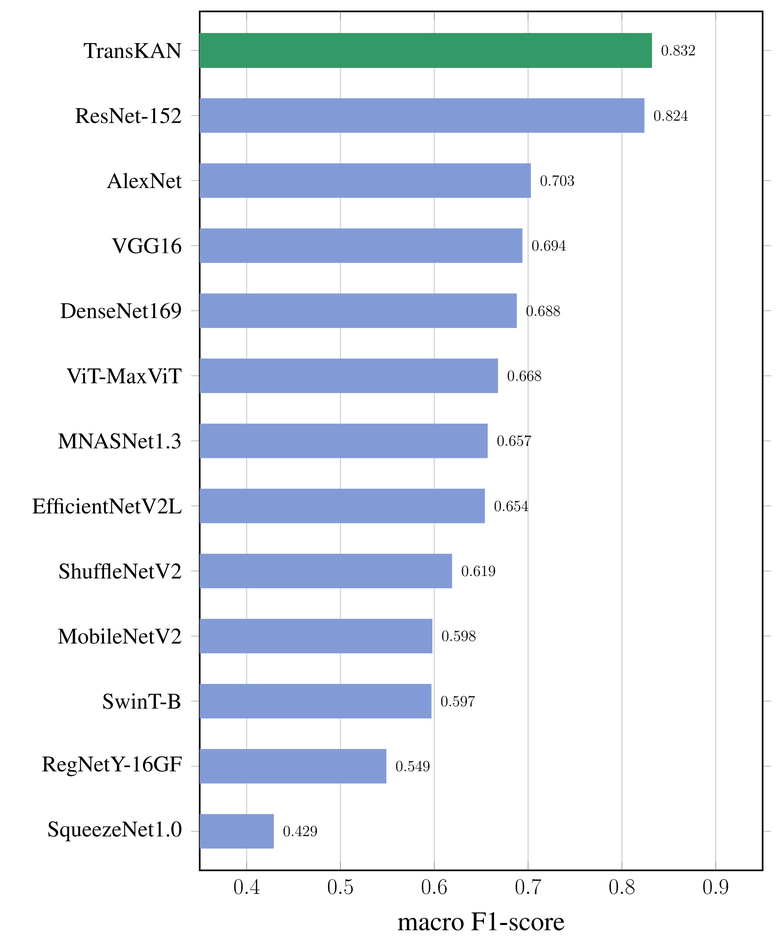


Figure 5: Wound-type classification, grouped 5-fold CV. Best variant of each architecture family. Lightweight models (MobileNet, ShuffleNet, SqueezeNet) form the lower bound. Isotropic ViTs are inconsistent at this data scale. Deep residual backbones are strongest, and **TransKAN matches or exceeds ResNet-152 on every primary metric**.

	Precision macro / micro	Recall macro / micro	F1 macro / micro	AUC macro / micro	Specificity macro / micro
ResNet-152	.827 / .869	.821 / .869	.824 / .869	.821 / .821	.821 / .869
TransKAN	**.847 / .871**	.821 / **.871**	**.832 / .871**	**.893 / .924**	.821 / **.871**

Table 1: Head-to-head with the strongest baseline (out-of-fold, grouped 5-fold CV). In this binary task, the micro averages equal accuracy.

Future Work

Joint multi-task entry/exit and MLSD (contact / close-range / distant) estimation. Wavelet-based KANs to cut cost. External validation on international forensic databases. Code and data will be released after acceptance.

References. [1] Z. Liu *et al.*, "KAN: Kolmogorov–Arnold Networks," ICLR 2025. [2] R. Queiroz Nogueira Lira *et al.*, "Deep learning-based human gunshot wounds classification," *Int. J. Legal Med.*, 139(2), 2025. [3] K. He *et al.*, "Deep residual learning for image recognition," CVPR 2016. [4] Efficient-KAN, github.com/Blealtan/efficient-kan. [5] A. Buslaev *et al.*, "Albumentations," *Information*, 11(2), 2020.

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